DEFENCE JOURNEY

NDA General Science Notes

Biology, environment and science and technology for the NDA GAT paper

NDA | SCIENCE

General Science in NDA includes Biology (life science), environment and basic science and technology. Questions are NCERT based (Class 6 to 12). These notes cover the topics that come up most often.

1. The Cell

- The cell was discovered by **Robert Hooke** (1665). Cell theory: Schleiden and Schwann. Cells arise from pre-existing cells: Rudolf Virchow.
- **Prokaryotes** (bacteria, blue-green algae) have no true nucleus. **Eukaryotes** (plants, animals, fungi) have a nucleus enclosed in a membrane.
- **Nucleus** controls the cell and carries DNA. **Mitochondria** are the powerhouse (make ATP). **Ribosomes** make proteins. **Golgi apparatus** packs and sends out substances.
- **Lysosomes** digest waste (suicide bags). **Endoplasmic reticulum** transports material. **Chloroplast** (plants only) does photosynthesis. **Vacuole** is large in plant cells.
- Only plant cells have a **cell wall** (made of cellulose) and chloroplasts. Animal cells have centrioles.
- Smallest cell: Mycoplasma (PPLO). Largest cell: ostrich egg. Longest cell: nerve cell. RBCs have no nucleus in humans.

2. Human Body Systems

System	Main organs	Key facts
Digestive	mouth, stomach, small and large intestine, liver, pancreas	Saliva (amylase) digests starch. Pepsin and HCl act in the stomach. Bile (liver) emulsifies fats. Digestion ends and absorption happens in the small intestine (villi). The large intestine absorbs water.
Circulatory	heart, arteries, veins, capillaries	Heart has 4 chambers and beats about 72 times per minute. Arteries carry blood away from the heart, veins carry blood to the heart. Normal blood pressure 120/80 mm Hg.
Respiratory	nose, trachea, lungs, diaphragm	Gas exchange happens in the alveoli. Inhaled air is about 21% oxygen. Haemoglobin carries oxygen.
Excretory	kidneys, ureters, bladder	The nephron is the functional unit. Kidneys filter blood and make urine. Dialysis is used in kidney failure.
Nervous	brain, spinal cord, nerves	Neuron is the unit. Cerebrum for thinking, cerebellum for balance, medulla for involuntary actions (heartbeat, breathing). Reflex actions are controlled by the spinal cord.
Endocrine	pituitary, thyroid, pancreas, adrenal, gonads	Hormones are chemical messengers. Pituitary is the master gland.
Skeletal	bones, joints, cartilage	An adult has 206 bones, a newborn about 300. Longest bone: femur. Smallest bone: stapes (ear). 32 teeth in an adult, 20 milk teeth in a child.

Blood

- Blood volume of an adult: about 5 to 6 litres. Blood has plasma, **RBCs** (carry oxygen, live about 120 days, made in bone marrow), **WBCs** (fight infection) and **platelets** (clotting).
- Blood groups A, B, AB and O were discovered by **Karl Landsteiner**. O negative is the universal donor, AB positive is the universal recipient. The Rh factor was found in rhesus monkeys.
- Haemoglobin contains iron. Low haemoglobin causes anaemia.

Hormones

Gland	Hormone	Function / disorder
Pituitary	Growth hormone	Growth. Too little: dwarfism. Too much: gigantism
Thyroid	Thyroxine	Metabolism. Iodine deficiency causes goitre
Pancreas	Insulin, glucagon	Controls blood sugar. Lack of insulin causes diabetes
Adrenal	Adrenaline	Fight or flight, raises heart rate
Testes / Ovary	Testosterone / Oestrogen	Male and female characters

3. Nutrition: Vitamins and Minerals

Vitamin	Source	Deficiency disease
A	carrot, papaya, liver, milk	Night blindness (xerophthalmia)
B_1 (thiamine)	whole grains, pulses	Beriberi
B_2 (riboflavin)	milk, eggs	Cracked lips, mouth sores
B_3 (niacin)	meat, groundnut	Pellagra
B_{12}	meat, fish, milk	Pernicious anaemia
C (ascorbic acid)	citrus fruits, amla, guava	Scurvy (bleeding gums)
D	sunlight, fish liver oil	Rickets in children, osteomalacia in adults
E	vegetable oils, nuts	Sterility (reduced fertility)
K	green leafy vegetables	Poor blood clotting

- Vitamins A, D, E and K are **fat-soluble**. Vitamins B and C are **water-soluble**.
- **Minerals:** calcium (bones and teeth), iron (haemoglobin, anaemia), iodine (thyroid, goitre), sodium and potassium (nerve function), fluoride (teeth).
- Protein deficiency: **kwashiorkor** and **marasmus** in children. A balanced diet has carbohydrates, proteins, fats, vitamins, minerals, fibre and water.
- Carbohydrates and fats give energy. Proteins build the body. 1 g carbohydrate or protein = 4 kcal, 1 g fat = 9 kcal.

4. Diseases

Cause	Diseases
Bacteria	Tuberculosis, cholera, typhoid, tetanus, diphtheria, whooping cough, plague, leprosy, pneumonia
Viruses	Polio, measles, mumps, chicken pox, influenza, rabies, dengue, hepatitis, AIDS (HIV), COVID-19
Protozoa	Malaria (Plasmodium, spread by the female Anopheles mosquito), kala-azar, amoebic dysentery, sleeping sickness
Fungi	Ringworm, athlete's foot
Worms	Elephantiasis (filaria), ascariasis, tapeworm infection
Not infectious	Diabetes, cancer, goitre, haemophilia, colour blindness (genetic)

- **Vaccines:** BCG (tuberculosis), DPT (diphtheria, whooping cough, tetanus), polio (Salk injectable, Sabin oral), MMR (measles, mumps, rubella).
- **Edward Jenner** developed the first vaccine (smallpox). **Louis Pasteur** developed the rabies vaccine. **Alexander Fleming** discovered penicillin (1928).
- Smallpox was declared eradicated in the world in 1980. India was declared polio-free by WHO in 2014.
- Dengue and chikungunya are spread by the Aedes mosquito. Malaria by the Anopheles mosquito. Plague by rat fleas.

- AIDS is caused by HIV, which attacks the immune system (T cells). It spreads through blood, unsafe sex and from mother to child. It does not spread by touch or by sharing food.

5. Plant Life

- **Photosynthesis:** $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$, using sunlight and chlorophyll in the chloroplasts. Oxygen released comes from water.
- **Transpiration** is loss of water vapour through the stomata. **Xylem** carries water and minerals upward. **Phloem** carries food.
- Plant hormones: **auxin** (growth, phototropism), **gibberellin** (stem growth), **cytokinin** (cell division), **ethylene** (fruit ripening), **abscisic acid** (stress, seed dormancy).
- Pollination is the transfer of pollen to the stigma. Fertilisation produces the seed, and the ovary becomes the fruit.
- Monocots have one seed leaf and parallel venation (wheat, maize). Dicots have two seed leaves and reticulate venation (pea, mango).
- Nitrogen fixation is done by bacteria such as Rhizobium in the root nodules of legumes (pulses). Fertile soil gets nitrogen this way.
- Insectivorous plants (pitcher plant, Venus flytrap, sundew) grow in nitrogen-poor soil. Parasitic plant example: Cuscuta (amarbel).

6. Genetics and Classification

- **Mendel** is the father of genetics (pea plant experiments). DNA's double-helix structure: Watson and Crick (1953).
- A human cell has **46 chromosomes (23 pairs)**. Sex chromosomes: XX in females, XY in males. The father's chromosome decides the sex of the child.
- Genes are units of heredity on the DNA. A change in the DNA is a **mutation**. Genetic disorders: haemophilia, colour blindness, sickle-cell anaemia, Down syndrome (extra chromosome 21).
- **Linnaeus** introduced binomial nomenclature (genus + species, e.g. *Homo sapiens*). **Whittaker's five kingdoms:** Monera, Protista, Fungi, Plantae, Animalia.
- Mammals are warm-blooded, have hair and feed milk to young. The bat is the only flying mammal. The whale and dolphin are mammals, not fish. Platypus is an egg-laying mammal.
- Amphibians (frog) live on land and water. Reptiles (snake, lizard) are cold-blooded with scales. Birds have feathers.

7. Environment and Ecology

- An **ecosystem** includes living (biotic) and non-living (abiotic) parts. Food chain: producers (plants) → herbivores → carnivores → decomposers.
- **10% law** (Lindeman): only about 10% of energy passes from one trophic level to the next.
- **Biodiversity hotspots in India:** Himalaya, Indo-Burma, Western Ghats and Sri Lanka, and Sundaland (Nicobar Islands).
- India's first national park: **Jim Corbett** (1936, Uttarakhand). Project Tiger started in 1973. Gir (Gujarat) is the home of the Asiatic lion. Kaziranga (Assam) is famous for the one-horned rhinoceros.

- **Ozone layer** in the stratosphere protects from UV rays. Montreal Protocol (1987) controls ozone-depleting substances. The Paris Agreement (2015) is for climate change.
- **Greenhouse gases:** CO_2, methane, nitrous oxide, CFCs, water vapour. Global warming is the rise of Earth's average temperature.
- **Acid rain** is caused by sulphur dioxide and nitrogen oxides. Smog is smoke + fog. Eutrophication is excess nutrients in a water body.
- Important days: Earth Day (22 April), World Environment Day (5 June), World Health Day (7 April), World Water Day (22 March).
- Environment laws in India: Wildlife Protection Act 1972, Water Act 1974, Air Act 1981, Environment Protection Act 1986.

8. Science and Technology in India

Area	Facts
Space	ISRO was formed in 1969 (HQ Bengaluru). Aryabhata (1975) was the first Indian satellite. Chandrayaan-1 (2008), Mars Orbiter Mission or Mangalyaan (launched 2013, reached Mars in 2014 in the first attempt), Chandrayaan-3 landed near the Moon's south pole on 23 August 2023.
First in space	Rakesh Sharma (1984, first Indian astronaut). Kalpana Chawla (first Indian-born woman in space, 1997).
Launchers	PSLV (workhorse), GSLV, and LVM3 (used for Chandrayaan-3).
Missiles (DRDO)	Agni (surface-to-surface, long range), Prithvi, Akash (surface-to-air), Nag (anti-tank), BrahMos (supersonic cruise missile, with Russia).
Nuclear	Pokhran tests in 1974 (Smiling Buddha) and 1998. India follows a no-first-use policy. BARC is at Trombay, Mumbai.
Computing and biotech	Param supercomputers (C-DAC). Internet, mobile networks and AI are growing areas. Biotechnology includes genetic engineering and vaccines.

TIP: Read the NCERT Science books of Class 6 to 10 once for Biology. Most questions in NDA are straight from these chapters.